

Daniel Simpson

CHEM 322 002

Sica, V. P.; Mitchell, C. A.; Krzykwa, J.; Baker, T. R.; Suramya Waidyanatha. *Analytical and Bioanalytical Chemistry*. **2025**, 417, 4895–4906.

This paper aims to quantitatively and qualitatively characterize the chemical profile of a botanical extract, using ashwagandha root extract for example. The use of non-nutritional botanical and herbal supplements is becoming more widespread in the United States, yet there's a lot of variability in botanical-derived products because of differences in their growth and processing. Especially because these supplements aren't FDA regulated, it's important for the safety of the general public to understand their composition.

The authors described a need for state-of-the-art analytical equipment because of the many constituents of the extract with different properties. The researchers chose ashwagandha because experts had prior analytical experience with the plant and its extract. They obtained one USP-grade sample of ashwagandha to use as a reference, and three extracts from three other suppliers recommended by Botanical Safety Consortium (BSC) experts. They used an ultra-high-performance liquid chromatography platform with photodiode array detection (PDA), charged aerosol detection (CAD), and high resolution mass spectrometry (MS), all of which complement each other, to ensure a comprehensive, holistic analysis.

The researchers prepared two standard solutions in 50:50 methanol and water with constituent concentrations ranging from 0.8 to 100 µg/mL, and triplicate ashwagandha samples at 20 mg/mL. The standards were prepared with analytes known to be found in ashwagandha. They also prepared the standards in two sets, staggering similar compounds to avoid co-elution; when

two peaks of the chromatogram overlap, making it harder to discern different compounds. The experiment did not prioritize speed; the researchers used longer, gradual chromatographic gradients and high sample concentrations to ensure maximal separation and to quantify as many constituents as possible.

A primary concern for the researchers would've been the quality and sourcing of the material to be analyzed which is why they used sources recommended by the BSC, and compared them to the USP-grade sample. Additionally, to compensate for changes in the composition of the 50:50 water and methanol that the extract was contained in, the researchers added a make-up flow of solvent before the system split into the CAD and mass spectrometer to maintain consistency in the ratio of the mobile phase. When the chromatographic sequence was completed, response factors were calculated for each standard at varying concentrations, for which the linearity was confirmed ($r^2 > 0.99$), and which were then averaged to create a relative response factor (RRF). The relative standard deviations of the individual response factors were confirmed to be below 25%, which was stated to be the reliable limit for this analysis. Using the RRF, each peak in the chromatogram could be integrated and the respective compound's concentration could be approximated. Additionally, the known values of compounds in the standards were compared with estimated values derived using the RRF. They were confirmed to be within 25% of each other, supporting the validity of the relative response factor.

The analysis resulted in 46 peaks that represented over 60 constituent substances. At the end of the characterization process, the researchers determined there to be 9 partial classifications (formula derived from mass measurement), 26 tentative (formula from mass measurement, and a proposed structure supported by MS/MS, UV-Vis, or literature data), 21 matched (confirmed via

mass spectrometry *and* existing literature data), and 6 reference classifications (confirmed based on compounds in their own standards). The researchers were able to conclude, with reasonable confidence, the composition by weight of the ashwagandha extract, which was tabulated in the paper. This data has the potential to be used to identify adulterants or contaminants and for toxicological risk assessment or comparison between botanicals from different sources. Knowing what is in the extract is the first step, further analyses can be performed if deemed necessary. This paper merely serves as an example of how to characterize the constituents of a substance of interest.